
Comparative Survival Modeling for Prediction of Incident Myocardial Infarction among ICU Patients

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Abstract

Intensive Care Unit (ICU) survivors represent a high risk population that is routinely discharged without individualized assessment of their cardiac risk post discharge. This study investigates whether structured baseline clinical data collected during the first 24 hours of ICU admission can predict time to first incident myocardial infarction (MI) among ICU patients without prior MI history. Using the MIMIC-IV database, a cohort of 33,072 patients with 660 post discharge MI events was constructed. Four survival models (Cox Proportional Hazards, Elastic Net Cox, Random Survival Forest, and Gradient Boosted Survival) were built and compared across C-index, AUC, and Brier Score metrics. Elastic Net Cox achieved the best overall performance with a 0.7008 C-index and 0.7765 1 year AUC where the Cox-based models consistently outperformed the tree-based models. Top predictors from the Cox-based models were well-established cardiovascular risk factors which supports the biological validity of the learned associations. Risk stratification using the best performing Elastic Net Cox model produced statistically significant separation between high and low risk groups. The results of the study demonstrate that routinely collected ICU data can meaningfully stratify post-discharge MI risk and support targeted cardiovascular follow-up at the time of discharge.

1 Introduction

Cardiovascular disease (CVD) is the leading cause of mortality worldwide, responsible for almost 20 million deaths in 2022 alone with approximately 85% of CVD-related deaths attributed to heart attack and stroke [1]. Myocardial infarction (MI), which is commonly known as a heart attack, occurs when blood flow to the heart is suddenly obstructed, leading to damage or death of heart tissue due to lack of oxygen. Majority of MI events also occur outside of the hospital setting where clinical intervention is not available [2].

Early risk identification is therefore critical. While comprehensive acute MI management strategies have improved survival following diagnosis, identifying individuals at elevated risk before an MI occurs remains a challenge. According to the World Health Organization, most MIs can be prevented by addressing behavioral and environmental risk factors such as smoking, diet, obesity, physical inactivity, alcohol use, and air pollution exposure [1]. Early detection would enable preventative measures before an irreversible cardiac event occurs. However, standard prevention frameworks such as the Framingham Risk Score were designed for the general outpatient population rather than for patients with already complex medical histories. Most ICU survivors are discharged without any individualized assessment of their post-discharge cardiac risk which represents an opportunity for targeted risk stratification [3].

2 Background

Prior work on cardiovascular risk prediction can generally be grouped into prediction in general outpatient populations and prognosis in patients already diagnosed with a cardiac condition.

Several studies have developed CVD risk prediction models for general registry-based populations. Christensen et al. [4] built a nationwide Danish registry-based model for 5-year risk of first CVD event using a composite outcome that includes ischemic heart disease, stroke, and heart failure. Liu et al. [5] benchmarked survival machine learning models for 10-year CVD risk prediction using a large-scale UK EHR database. While results demonstrate the feasibility of machine learning based time-to-event prediction, their target population was primary care patients aged 40 to 85. Mandair et al. [6] directly addressed incident MI prediction while using EHR data, but it was framed as a 6 month classification problem rather than a censored time-to-event problem. The authors also found that a deep neural network achieved the best classification performance, but improvement over the logistic regression model using known MI risk factors was modest.

Furthermore, existing literature also largely focuses on patients who have already experienced a cardiac event [7, 8, 9, 10, 11]. Lin et al. [7] used MIMIC-IV to predict 30-day mortality among critically ill patients already diagnosed with MI and develop an ML-assisted nomogram (visual clinical prediction tool) that outperformed SOFA in discrimination. However, their outcome was short-term mortality among MI patients admitted to the CCU. Hadanny et al. [9] similarly predicted 1 year mortality following acute coronary syndrome using Random Survival Forest, DeepSurv, and Cox Proportional Hazards models, while Farhadian et al. [10] compared survival models to predict 10-year major adverse cardiovascular and cerebrovascular events (MACCE) after percutaneous coronary intervention (PCI). While these studies demonstrate the effectiveness of survival and machine learning methods to study cardiac populations, they address prognosis after established cardiac disease or procedure-specific outcomes rather than pre-event risk prediction in ICU survivors.

A systematic review of AI models for time-to-event CVD prediction also found that Random Survival Forest, Gradient Boosted Survival, penalized Cox models, and DeepSurv were among the most frequently used approaches. It also emphasized that survival outcomes require modeling both follow-up time and event status [12].

Overall, prior work demonstrates that cardiovascular risk prediction can be meaningfully improved using large-scale EHR data and survival machine learning models as EHR-derived datasets can capture "large, heterogeneous, and contemporaneous patient populations" [13]. However, existing studies either focus on broad composite CVD outcomes in general populations, short term mortality among patients already diagnosed with a cardiac condition, or procedure specific outcomes. The specific task of predicting time to first post-discharge MI among ICU survivors without MI history at index admission remains largely unaddressed.

3 Clinical Motivation

The classical approach to CVD prevention relies on tools such as the Framingham Risk Score which estimates 10-year CVD risk from outpatient measurements such as cholesterol, blood pressure, age, sex, and smoking status [14]. Such tools are useful for the general population, but they were not designed for patients with complex medical histories such as post-ICU survivors. Patients discharged from the ICU following sepsis, respiratory failure, or acute kidney injury are physiologically and clinically different from a typical patient visiting their primary care doctor for an annual check up, and standard risk calculators do not account for this disparity. The 10 year prediction horizon for the Framingham tool is intentional because it motivates interventions and lifestyle changes that require years to accumulate meaningful reduction in cardiovascular risk. Post-ICU patients however are already medically complex where a lifestyle change alone is not the appropriate clinical intervention. Instead, they would require clinical surveillance and care coordination such as prioritized cardiology referral, earlier secondary prevention, patient education about warning signs, and scheduled follow-ups.

Most ICU survivors, however, are discharged without any individualized assessment of their post-discharge cardiac risk [3]. Thus, the goal of this project is to determine whether structured clinical data generated during their ICU stay can predict the time-to-first incident MI after discharge. Estimating patient risk levels can inform a risk stratification tool built specifically for medically complex, post-

ICU patients who require a different kind of clinical response than lifestyle changes alone. This study would help clinicians determine patients who are at elevated risk and warrant prioritized follow up before leaving the hospital.

4 Methods

4.1 Study Design

This study uses the MIMIC-IV dataset which is comprised of ICU admissions from 2008 to 2019 at Beth Israel Deaconess Medical Center. This dataset was selected primarily due to data availability, but it also consists of complete longitudinal inpatient records for each ICU admission. It captures valuable data on vital signs, lab panels, severity scores, medications, and intervention records that describe the patient’s physiological state which allows for a robust construction of a patient cohort.

Additionally, survival analysis is the most appropriate modeling framework for time-to-event problems. Binary classification models require a fixed prediction horizon and do not account for censoring. Survival analysis has the ability to handle right-censored observations and produces horizon-flexible risk estimates. Cox models also yield interpretable hazard ratios that can directly quantify which clinical features are associated with an increased hazard for MI.

We, therefore, hypothesize that baseline ICU clinical features (including demographics, vitals, lab values, severity scores, and comorbidities) can predict hazard of post-discharge incident MI among ICU survivors without prior MI history. Survival models will achieve meaningful discrimination with a C-index greater than 0.65, while clinically recognized MI risk factors such as age, diabetes, and comorbidity burden will emerge as top predictors. Additionally, predicted risk scores will produce statistically significant stratification between high and low risk patient groups to support the utility of the model as a discharge-time risk stratification tool.

Patient Cohort: The patient cohort included adult patients (age ≥ 18) with a first ICU admission, no prior or index MI history, valid discharge timestamp, first 24-hour labs and vitals available, and no patient death during the index admission. Features were restricted to the first 24 hours of ICU admission because it represents the earliest and most complete snapshot of a patient’s physiological state where clinicians have ordered labs and established vital sign monitoring. Restricting to a fixed window also standardizes the feature set across all of the patients regardless of ICU length of stay. This ensures that differences in predicted risk reflect differences in clinical characteristics rather than differences in how long a patient was observed for.

The final cohort contained 33,072 patients with a median age of approximately 63 years. The racial distribution was predominantly White at 64.3% followed by Black at 7.3% and Hispanic at 3.4%. The median Charlson Comorbidity Index was 3 which reflects a higher burden of chronic disease in ICU populations, and the median follow-up for censored patients was around 3 to 5 years with some patient follow-ups ranging over 14 years long. There were 660 MI events in total which yields a low event rate of approximately 2%. Among the censored patients, 19,343 patients had no recorded post-discharge contact or death within MIMIC-IV which accounts for around 58% of the entire cohort. Among patients who did experience a post-discharge MI, the median time-to-event was 780 days. This suggests that MI is a longer horizon outcome rather than an immediate event following ICU discharge.

Features: Using Google BigQuery, information was collected from the MIMIC dataset to be used as features and were grouped into demographics, healthcare utilization, vital signs and lab values from the first 24 hours, severity scores, charlson comorbidities, additional comorbidities, medications taken at index admission, and ICU interventions. A comprehensive list of all the features used post-processing can be found in Appendix A.

The time origin was hospital discharge from the index ICU admission, while the outcome was time to first post-discharge MI and was defined as the number of days from hospital discharge to the first subsequent admission carrying an acute MI diagnosis. Patients who passed away without experiencing a post-discharge MI and patients with no additional follow up after their first ICU discharge were right-censored at their last known contact date. Across this patient cohort, the length of follow-up ranged from 1 day to over 14 years. Patients with 0 follow up days due to right censoring were clipped to 1 day rather than being excluded entirely.

4.2 Data Preprocessing

The raw feature matrix contained 106 columns with an overall missingness rate of 14.37%. To handle missing values, numeric features were categorized and handled using a tiered strategy based on missingness severity.

Features exceeding 75% missingness were removed entirely from the feature set, while features with 20 to 75% missingness were retained using median imputation. A binary `_measured` indicator was created for each of these features before imputation that records whether a value was observed or not. Features in this range were predominantly laboratory values, which are measurements selectively ordered based on clinician judgment. Whether a clinician chose to order a particular test may carry prognostic signal. Features with less than 20% missingness were imputed based on their skew distribution. Using the Fisher-Pearson method, features with absolute skewness exceeding 1 indicate a meaningfully skewed distribution and were imputed using the median. Any features with absolute skewness at or below 1 were imputed using the mean.

All categorical features with low missingness were handled separately from the numeric features. Those with low missingness were imputed using the mode while those with higher missingness were assigned an "Unknown" category to preserve the value of absent information. Binary indicator flags with zero variance after imputation were dropped. Additionally, outliers in continuous features were handled using percentile-based Winsorization which caps values at the 1st and 99th percentiles to remove any physiologically implausible values or data entry errors while preserving the overall shape of the distribution.

4.3 Feature Engineering

For vital signs, two representations were used, the 24-hour mean and a clinically meaningful extreme. The mean captures the patient's typical state over the 24-hour observation window, while the extreme captures the worst physiological deviation. The direction of the extreme (minimum or maximum) was selected based on clinical risk. For example, minimum was considered most appropriate for representing blood pressure and oxygen saturation where hypotension and hypoxemia represent dangerous deviations. Maximum was chosen for heart rate and respiratory rate where tachycardia and tachypnea signal physiological stress.

Conducting a Pearson correlation analysis on all continuous features, all pairs with an absolute correlation value greater than or equal to 0.90 were flagged. Any redundant features were removed based on additional clinical review. If any pair of features captured the same underlying construct, the less clinically precise or more derived variable was dropped. For example, `hemoglobin_min` was removed in favor of `hematocrit_min` as correlated CBC markers while the liver function marker `ast_max` was removed in favor of `alt_max`. Features that reflect the same underlying disease process as MI, such as heart failure and atrial fibrillation/flutter, were also excluded to prevent the model from predicting MI using features that are symptoms or consequences of the same condition. Missingness indicator flags within the same laboratory panel were consolidated into a single panel-level indicator because tests within the same panel are often ordered together and therefore produce near perfectly correlated missingness patterns.

Race/ethnicity string values were consolidated into seven categories: White, Black, Hispanic, Asian, Native American, Pacific Islander, and Unknown/Other. All categorical features were one-hot encoded with reference category drop to avoid multicollinearity.

The final feature matrix contained 87 predictors prior to encoding (Appendix A). To account for the low event rate of 2%, the dataset was split 80-20 into training and test sets of sizes 26,457 and 6,615. This split was performed prior to any scaling by using stratified sampling on the event indicator to preserve the event rate in both sets. To prevent data leakage, the scaler was fit only on the training set before applying the training derived scaling parameters to the test set. Afterwards, all continuous features were standardized to zero mean and unit variance, and all binary features and missingness flags were left unscaled.

4.4 Model Development and Validation

Four survival models were developed including both Cox-based and tree-based methods. Cox Proportional Hazards and Elastic Net Cox were included for their interpretability and wide use in

clinical risk prediction studies. Tree-based ensemble methods such as Gradient Boosted Survival and Random Survival Forest were selected to test whether nonlinear feature interactions could improve discrimination. Additional details on model definitions and specifications are in Appendix B.

All four models were tuned using 3-fold cross validation on the training set with optimizing the C-index as the tuning objective. In other words, the average C-index across all of the folds was used to select the optimal hyperparameter configuration. For Cox Proportional Hazards, a grid search was conducted over the penalizer strength with L1 ratio fixed at 0. For Elastic Net Cox, the regularization path was first fit on the full training data across 100 alpha values. Rather than evaluating all 100 alphas via cross validation, every 10th alpha was selected, yielding approximately 10 candidate values spread across the full penalty range. 3-fold CV was then used to run on these candidate alphas, and the one that maximized mean C-index across folds was selected as the final regularization strength. For Random Survival Forest, grid search was conducted over number of estimators and minimum samples per leaf. For Gradient Boosted Survival, grid search covered number of estimators, learning rate, maximum depth, and minimum samples per leaf. The final tuned hyperparameters for each model are reported in Table 1 and additional information about the hyperparameter tuning process can be found in Appendix D.

Table 1: Hyperparameter Tuning Results

Model	Tuned Hyperparameters	CV C-Index ($\pm$ SD)	Test C-Index
Cox PH	penalizer=0.01, l1_ratio=0.0	0.6916 ± 0.0222	0.6976
Elastic Net Cox	l1_ratio=0.5, alpha=0.001229	0.6972 ± 0.0313	0.7008
Random Survival Forest	n_estimators=200, min_samples_leaf=10, max_features="sqrt"	0.67 ± 0.02	0.7002
Gradient Boosted Survival	learning_rate=0.10, max_depth=3, n_estimators=150, subsample=0.8	0.6753 ± 0.0016	0.6860

*Best-performing value among compared models for the corresponding metric.

4.5 Evaluation Metrics

To measure discrimination, Concordance index (C-index) and time-dependent AUC were computed for each of the four models, and time-dependent AUC was evaluated at specific time horizons of 1, 2, and 5 years. To measure calibration, Brier Score at 1 year and Integrated Brier Score (IBS) were used as well. For more details on these metrics, please see Appendix B.

5 Model Performance and Results

Based on the performance of the four models as shown in Table 2, Elastic Net Cox achieved the highest C-index of 0.7008 and Brier Score of 0.0252 while Cox Proportional Hazards achieved the highest discrimination at 1 and 2 year horizons with AUC scores of 0.780 and 0.8008 respectively. Random Survival Forest performed comparably to both Cox-based models with C-index but had lower time-dependent AUC scores across all three horizons. Gradient Boosted Survival, however, consistently underperformed relative to the three other models across all metrics. Overall, all four models exceed the initial hypothesis thresholds of a C-index greater than 0.65.

The most important finding from these performance metrics is that the Cox-based models outperformed the tree-based models. This suggests that the relationship between the ICU features and post-discharge MI hazard is mostly additive and linear on the log-hazard scale. This is consistent with how cardiovascular risk is understood clinically where established risk factors like age and diabetes each independently elevate risk in a predictable direction. With only 600 MI events out of 33,072 patients, there is relatively little data for the model to learn from compared to the number of features

used. Tree-based models can detect nonlinear relationships between features, but with a low event rate, the model may learn patterns that are specific to the training data but do not hold for patients it has not seen before such as with the test set.

Table 2: Model Performance Comparison on Test Set

Model	C-Index	AUC (1 yr)	AUC (2 yr)	AUC (5 yr)	Brier (1 yr)	IBS
Cox Proportional Hazards	0.6976	0.7870	0.8008	0.7418	0.0256	0.0487
Elastic Net Cox	0.7008*	0.7765	0.7895	0.7496*	0.0252*	0.0478*
Random Survival Forest	0.7002	0.7656	0.7816	0.7278	0.0255	0.0481
Gradient Boosted Survival	0.6860	0.7371	0.7698	0.7315	0.0258	0.0498

Using the Elastic Net Cox model, patients were stratified into high risk and low risk groups using a 75th percentile threshold. The Kaplan-Meier curves in Figure 1 show visually clear separation between the two risk groups. A log rank test was conducted with a p-value of 6.39×10^{-9} which confirms that there is indeed statistical significance between the two groups. It can also be seen that the high risk group maintains around 70 to 75% MI-free survival at around 3000 days, or 8 years, post-discharge. Confidence intervals also widen substantially beyond 4000 days due to sparse event counts at longer follow up times. The separation between the groups however remains significantly different throughout the observation period.

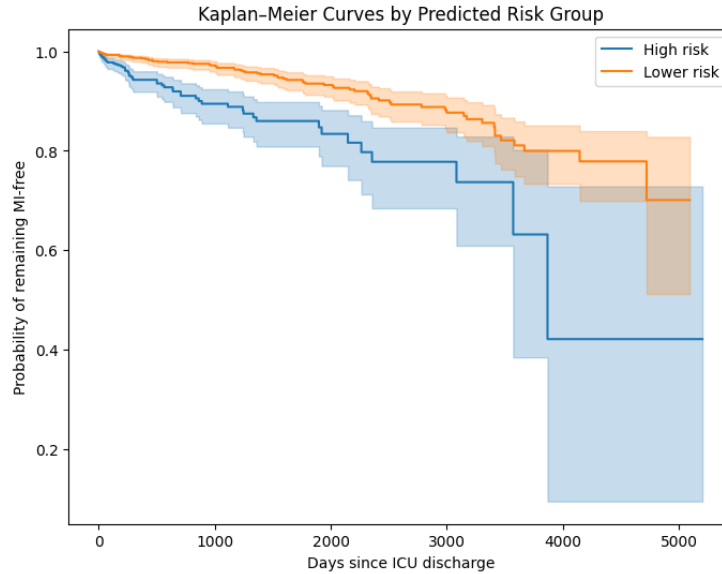


Figure 1: Kaplan-Meier Survival Curve by Risk Group

The calibration curve at 365 days plots each model’s mean predicted 1-year MI risk against the observed 1-year event rate across risk bins where the dashed line represents perfect calibration (i.e. predicted risk = measured risk). It can be seen in Figure 2 that all four models cluster their predictions in a narrow range between 0 and 0.03 predicted risk which is consistent with the overall 2% event rate in the cohort. All models also lie close to the diagonal at low predicted risk levels which suggests adequate calibration in the majority of patients who carry lower risk. As predicted risk increases beyond 0.02, the models start to diverge away from the diagonal.

While all models generally capture the average event rate, none of them achieve well-calibrated absolute risk estimates across the full range of predicted probabilities. Therefore, post-hoc recalibration is likely needed, such as isotonic regression, on the predicted probabilities before these values can be used for individual level risk quantification in a clinical setting. For more details on the calibration analysis as well as reported risk ranges for each model, see Appendix E.

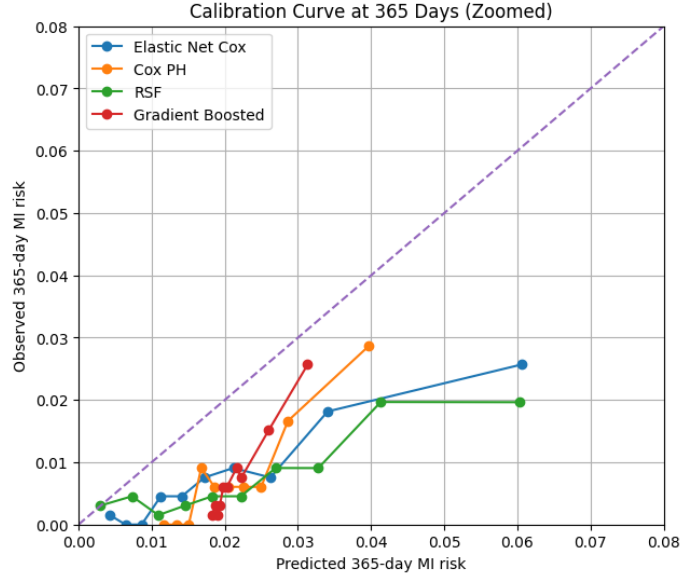


Figure 2: Calibration at 365 Days

Moreover, the top six statistically significant predictors from the Cox Proportional Hazards model were diabetes_with_cc, ckd (chronic kidney disease), diabetes, charlson_comorbidity_index, age, and bun_max (Appendix F). These are all considered clinically well-established MI risk factors where diabetes with complications carried the largest hazard ratio of 1.24. CKD and uncomplicated diabetes both independently contribute to MI hazard beyond the overall Charlson Comorbidity Index, while age and BUN max show tighter confidence intervals. This reflects more significant signal across the full patient population. All six predictors also have hazard ratios above 1.0 with confidence intervals that confirm statistically significant positive associations with MI hazard.

The Elastic Net Cox Model selected a sparse set of predictors with non-zero coefficients at the optimal alpha (Appendix G). The top predictors were largely consistent with the Cox Proportional Hazard model's results where Charlson comorbidity index and age were the two strongest signals. Additions that were not seen in the Cox Proportional Hazard results include male sex and an admission type indicator with hazard ratios of 1.25 and 1.17 respectively. These results show that the clinical context of the index ICU admission carries independent prognostic information beyond the patient's baseline medical profile. Additionally, diuretic use resulted in a hazard ratio of 1.18 but is likely a confounding factor that acts as a proxy for underlying cardiovascular burden rather than a direct causal risk factor. It is also worth pointing out that length of stay independently predicts MI risk with a hazard ratio of 1.10, suggesting that patients with longer and more complicated ICU stays carry higher downstream cardiac risk.

5.1 Subgroup Analysis

A subgroup analysis was conducted to better understand the model's performance with respect to sex and age. The sex subgroup analysis showed near-identical discrimination with AUC scores of 0.78 and 0.78 for male and females respectively. This suggests that the model generalizes comparably across sexes. However, the age subgroup results in a larger performance gap where patients under 65 achieved AUC of 0.84 compared to 0.66 observed in patients aged 65 and above. However, this AUC difference may not necessarily reflect model bias and instead be likely due to the inherent difficulty of predicting MI in an older ICU population. Older patients tend to have more complex comorbidity profiles which makes risk less separable, and they also face higher competing risks from non-MI related mortalities such as cancer and dementia. See Appendix H for the subgroup analysis numbers.

6 Limitations and Challenges

The primary limitation of this study is the low event rate of approximately 2% which creates several challenges. Statistical estimates involve wider confidence intervals that limit interpretability, and tree-based models struggle with extreme class imbalance. As seen in Figure 2, calibration at high predicted probabilities is unreliable due to sparse event counts for high risk bins. The low event rate also reflects a deliberate design choice to preserve MI as a standalone outcome rather than collapsing it into a broader composite MACE outcome. While the incorporation of synthetic data may be a possible next step, a model trained on artificially inflated event counts would likely learn a different risk distribution and may not generalize to more realistic clinical settings.

Another limitation is that predictors were restricted to the first 24 hours of the index ICU admission. While this supports early prediction, the choice of this time window was not formally tested. Thus, future work should compare the 24 hour window to 48 and 72 hour windows to determine whether longer observation periods can better capture clinically meaningful physiological trajectories among patients. Furthermore, generalizability is limited by the use of a single dataset with model performance varying across hospitals and patient populations. The use of additional datasets to test the models' performance would better validate the study's results. There may also be unmeasured clinical factors that were not addressed in the study that could lead to residual confounding such as medication adherence, lifestyle factors (physical activity, alcohol use), and socioeconomic factors (access to follow up care, health literacy).

Future extensions of the study should also consider a competing risks framework with death without MI as a competing event instead of right censoring. This could improve risk estimation as death prevents the later occurrence of post-discharge MI. In addition, the current feature set summarizes early ICU measurements into fixed baseline variables such as the minimum or mean values from the first 24 hours. This approach captures overall severity but may miss patterns such as rapid deterioration or recovery in patients. Time-varying ICU covariates would therefore improve modeling of a patient's changing physiology over the course of their ICU hospitalization. Finally, unstructured data, such as clinical notes and imaging, may provide additional prognostic information beyond structured variables alone. It would also allow for the use of neural networks which opens up an area to explore this space further.

7 Conclusion

This study demonstrates that structured baseline ICU features collected during the first 24 hours of admission can stratify the risk of post-discharge incident MI in ICU survivors without prior MI history. The Elastic Net model achieved the best overall performance with a C-index of 0.7008 and 1-year AUC of 0.7765 with Cox-based models consistently outperforming tree-based models across all performance metrics. Top predictors from the Cox-based models are clinically well-established MI risk factors, and risk stratification using the Elastic Net Cox model produced statistically significant separation between high and low risk groups. The results of this study show that risk level identification prior to ICU discharge can lead to meaningful improvements in post-discharge cardiac care. As a result, patients identified as high risk can be prioritized for cardiology referral, secondary prevention medication review, and earlier outpatient follow-up.

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Appendix

A ICD Codes and Comprehensive Feature List

Acute MI was identified using the ICD-9 code 410x and ICD-10 codes I21x and I22x. These capture ST-elevation MI, non-ST-elevation MI, and subsequent MI. Acute MI codes were used to identify post-discharge MI events in subsequent ICU admissions.

Prior MI history was identified using ICD-9 code 412 and ICD-10 code I252 which were both documented in the data as "old myocardial infarction" and indicate a healed or previously documented MI. Patients carrying any prior MI history code at any admission prior to the index were excluded from the cohort.

Feature List:

Table 3: Final Feature List After Preprocessing

Category	Features
Demographics	age, gender, race, marital_status, insurance, admission_type, discharge_location, los_days
Healthcare Utilization	days_since_last_admission, num_prior_icu_stays
Vital Signs (first 24h)	heart_rate_mean, heart_rate_max, sbp_mean, sbp_min, dbp_min, mbp_mean, mbp_min, resp_rate_mean, resp_rate_max, temperature_mean, temperature_max, spo2_mean, spo2_min
Laboratory Values (first 24h)	hematocrit_min, platelets_min, wbc_max, bun_max, creatinine_max, glucose_max, sodium_min, potassium_max, bicarbonate_min, chloride_min, calcium_min, aniongap_max, albumin_min, bilirubin_total_max, inr_max, ptt_max, d_dimer_max, fibrinogen_min, alt_max, alp_max, abs_neutrophils_max, abs_lymphocytes_min, abs_monocytes_max, bands_max
Arterial Blood Gas (first 24h)	lactate_max, ph_min, po2_min, pco2_max, pao2fio2ratio_min, baseexcess_min
Anthropometrics	height, weight_admit, bmi_admit
Severity Scores	sofa_score, sapsii, gcs_min, gcs_motor, gcs_verbal, gcs_eyes
Charlson Comorbidities	charlson_comorbidity_index, diabetes_with_cc, cerebrovascular_disease, dementia, chronic_pulmonary_disease, rheumatic_disease, peptic_ulcer_disease, mild_liver_disease, severe_liver_disease, malignant_cancer, metastatic_solid_tumor
Additional Comorbidities	hypertension, diabetes, hyperlipidemia, ckd, smoking, obesity, prior_stroke, pvd
Medications	med_insulin, med_diuretic
ICU Interventions	mechanical_ventilation, vasopressor_use, renal_replacement_therapy

Please see bigquery.txt for the query used to pull the desired data from the MIMIC-IV database using Google BigQuery. Note that not all of the information pulled through BigQuery was used in the final feature set.

B Evaluation Metric Definitions

To measure discrimination, Concordance index (C-index) and time-dependent AUC were computed for each of the four models. C-index is the probability that, for a randomly selected pair of patients where one experienced MI first, the model correctly ranks the higher risk patient as having the earlier event. A C-index of 1.0 is perfect while 0.5 would be random. Time-dependent AUC is AUC evaluated at specific time horizons of 1, 2, and 5 years. It specifically measures whether the model correctly ranks cases (patients who experience MI by time t) above controls (patients still event-free at time t).

To measure calibration, Brier Score and Integrated Brier Score (IBS) were used. Evaluated at 1 year, Brier Score is the mean squared difference between predicted survival probability and the actual outcome at a specific time. Integrated Brier Score is simply the Brier Score integrated over the full follow-up period to provide a summary of calibration across all time points.

C Model Development Specifications

The Cox Proportional Hazards model estimates the hazard function as:

$$h(t|X) = h_0(t) * \exp(\beta_1 X_1 + \dots + \beta_p X_p)$$

where $h_0(t)$ is the baseline hazard and $\exp(\beta_i)$ is the hazard ratio associated with one-unit increase in feature X_i . The key assumption with this model is proportional hazards where the relative risk between any two patients remains constant over time.

The Elastic Net Cox model extends Cox Proportional Hazards by adding both L1 (Lasso) and L2 (Ridge) regularization. The L1 penalty shrinks coefficients of non-significant features to zero which enables sparsity and feature selection. The L2 penalty helps stabilize estimation when predictors are correlated.

The Random Survival Forest model extends the random forest ensemble to survival outcomes where an ensemble of survival trees is built on bootstrap samples of the training data. Each tree considers a random subset of features at each split before the individual tree predictions are aggregated and averaged across the ensemble. This method is non-parametric and does not incorporate the proportional hazards assumption, naturally capturing the nonlinear relationships and interactions between the different features.

The Gradient Boosted Survival model applies gradient boosting to the Cox partial likelihood. Decision trees are sequentially fitted based on the residual errors of the previous ensemble, with a learning rate controlling the size of each update. Because predictions are generated through a non-parametric tree ensemble, this model is not constrained by the proportional hazards assumption.

The Cox-based models using the lifelines and scikit-survival packages in Python required standardized continuous features, while the tree-based models using scikit-survival were fitted on the unscaled imputed feature matrix.

D Hyperparameter Tuning

Due to time constraints, the search space was quite narrow which is another limitation of the study. An additional next step would be to expand the grid search further for more robust hyperparameter tuning.

Table 4: Hyperparameter Grid Search Space

Model	Parameter	Values Searched	Fixed Parameters
Cox PH	penalizer	[0.001, 0.01, 0.1, 1.0]	l1_ratio = 0.0
Elastic Net Cox	alpha	every 10th value from path of 100 (n_alphas=100, alpha_min_ratio=0.01)	l1_ratio = 0.5 alpha_min_ratio = 0.01
RSF	n_estimators min_samples_leaf	[100, 200] [10, 20]	max_features = "sqrt" min_samples_split = 20
GBS	n_estimators learning_rate max_depth min_samples_leaf	[100, 150] [0.05, 0.1] [2, 3] [10, 20]	subsample = 0.8 min_samples_split = 20 loss = "coxph"

E Risk Ranges from Calibration Results

Table 5 displays the risk ranges and mean risk for each model. It can be seen that all four means cluster around 2% which suggests that all models are broadly calibrated. Gradient Boosted has the narrowest range and does not predict below 1.7% for any patient, while Cox Proportional Hazards has a moderate range. Random Survival Forest has a range that includes predictions of as low as 0% and as high as 13%. Elastic Net Cox, however, has the widest range out of the four models and appears to be the most aggressive in terms of separating low and high risk patients. Some patients are predicted to have a 24% 1-year MI risk. This result aligns with Elastic Net Cox receiving the best C-index. The wide upper tail of both Elastic Net Cox and Random Survival Forest suggests that high predicted probability bins contain very few patients. This makes observed rates in these bins statistically unreliable and which is discussed in the limitations section.

Table 5: Predicted Risk Distribution by Model at 365 Days

Model	Minimum Risk	Maximum Risk	Mean Risk
Elastic Net Cox	0.0002	0.2360	0.0204
Cox PH	0.0009	0.0810	0.0212
RSF	0.0000	0.1308	0.0237
Gradient Boosted	0.0168	0.0683	0.0217

F Top Predictors and Hazard Ratios from Cox Proportional Hazards

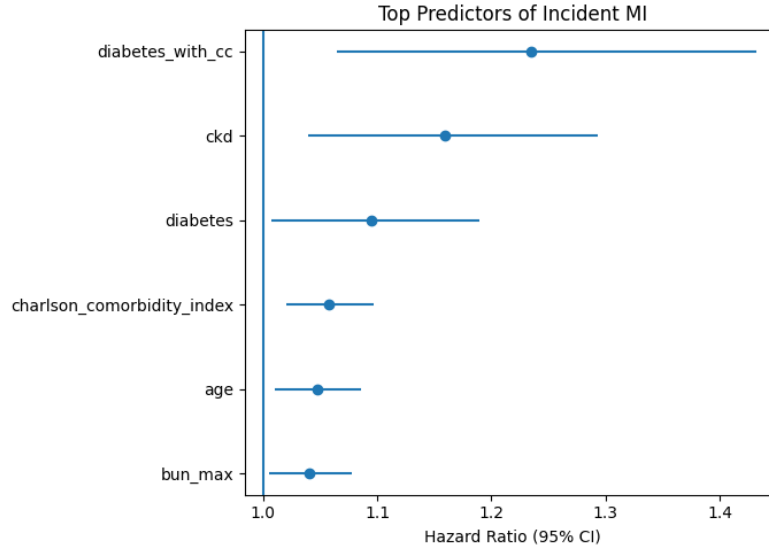


Figure 3: Top Predictors from Cox Proportional Hazards Model

Table 6: Top Cox PH Predictors with Confidence Intervals

Covariate	Coef.	HR	HR Lower 95%	HR Upper 95%	p-value
diabetes_with_cc	0.21	1.23	1.06	1.43	0.01
ckd	0.15	1.16	1.04	1.29	0.01
diabetes	0.09	1.09	1.01	1.19	0.03
charlson_comorbidity_index	0.06	1.06	1.02	1.10	< 0.001
age	0.05	1.05	1.01	1.09	0.01
bun_max	0.04	1.04	1.01	1.08	0.02

It is worth noting that the hazard ratio for diabetes with complications (diabetes_with_cc) is higher than that of uncomplicated diabetes (diabetes). Diabetes with complications is formally defined as the presence of end organ damage such as diabetic nephropathy or neuropathy. It is a disease state with greater cardiovascular risk than uncomplicated diabetes. Thus, the wider confidence interval for diabetes_with_cc relative to diabetes reflects the smaller number of patients who carry this more severe classification. Both features independently retaining statistical significance imply that diabetes and its complication status carry prognostic information beyond what the composite comorbidity score alone captures.

G Top Predictors from Elastic Net Cox Model

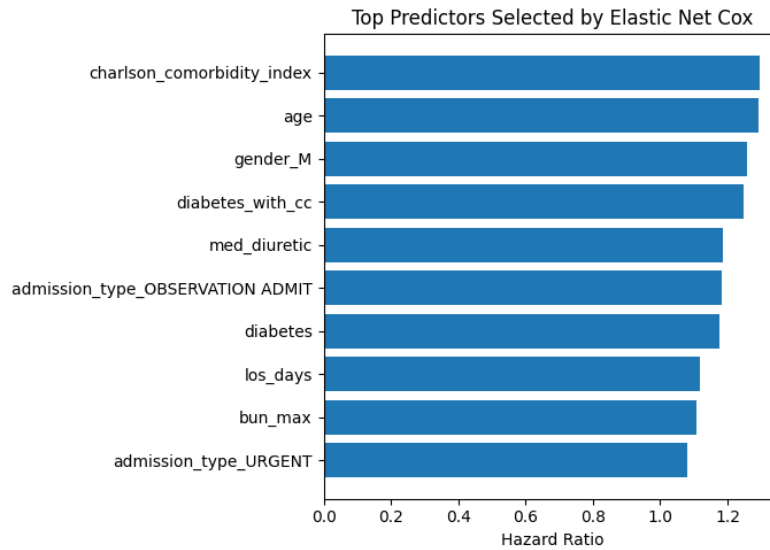


Figure 4: Top Predictors from Elastic Net Model

Table 7: Top Elastic Net Cox Predictors with Estimated Hazard Ratio

Feature	Coefficient	Hazard Ratio
charlson_comorbidity_index	0.26	1.30
age	0.26	1.29
gender_M	0.23	1.26
diabetes_with_cc	0.22	1.25
med_diuretic	0.17	1.19
admission_type_OBSERVATION ADMIT	0.17	1.18
diabetes	0.16	1.18
los_days	0.11	1.12
bun_max	0.10	1.11
admission_type_URGENT	0.08	1.08

The addition of "observation admit" (HR: 1.17) and "urgent admission (HR: 1.07) for the Elastic Net Cox model suggests that the clinical context of why a patient entered the hospital carries independent prognostic information beyond their baseline profile. Observation admissions are patients who were not formally admitted but were kept for monitoring. It is likely that these patients may represent an under-recognized high risk group who are discharged without the same level of follow up as formally admitted patients.

Length of stay is also clinically meaningful since longer ICU stays reflect greater illness severity and interventions. This may also lead to more physiological stress on the patient's cardiovascular system.

H Subgroup Performance Results

Table 8: Subgroup Analysis Results using Elastic Net Cox (Test Set)

Subgroup	N	Events	AUC (1 yr)	95% CI
Age < 65	3,511	48	0.84	[0.72, 0.91]
Age ≥ 65	3,104	84	0.66	[0.57, 0.75]
Female	2,920	55	0.78	[0.70, 0.85]
Male	3,695	77	0.78	[0.69, 0.86]